

DUET: Endogenous Failure Detection and Recovery via Dual-Action Manifold Alignment

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Abstract—Vision-language-action (VLA) models map vision and language to robot actions. With limited post-training demonstrations, however, they often cannot detect or recover from execution failures that move the robot out of distribution (OOD), such as mis-grasps, dropped objects, unexpected occlusions, and object or robot displacement. Existing solutions rely on external failure detectors and costly intervention or recovery data, leaving detection loosely coupled with control. We introduce DUET (Dual-manifold Unified Execution and Transfer), a dual-action-head module that predicts relative and absolute actions. DUET uses the endogenous disagreement between the two heads as its primary failure signal, complemented by state-distribution distance. The absolute head excludes the current robot state and predicts RVQ-anchored global waypoints from vision and language, providing stable recovery targets when relative control becomes unreliable. We evaluate DUET on four LIBERO suites under five families of execution failures, and transfer the identical module to a π_0 backbone on a real Franka arm. DUET preserves nominal success (96.9% vs. 93.8% for the relative-only baseline) while raising success under action-execution failures from 27.9% to 64.6% (+36.7 pp). The same disagreement score leads all success-only detectors on AUROC (0.845).

I. INTRODUCTION

Vision-language-action (VLA) policies adapt pretrained vision-language models to map visual observations and language instructions directly to robot actions [1], [2], [3]. Despite their semantic generalization, policies post-trained on limited robot demonstrations remain vulnerable to failures during closed-loop execution. We focus on failures that drive the visual observation, robot-object configuration, or induced trajectory out of distribution (OOD) relative to successful demonstrations. They include policy-induced errors, such as an off-center grasp or an accidentally dropped object, as well as external events, such as transient occlusion or unexpected displacement of an object or the robot arm. After such events, a VLA may continue to produce locally plausible actions even though its trajectory is no longer supported by the demonstrations. Detecting these failures and actively recovering from them can therefore substantially improve task success and deployment reliability.

Existing approaches commonly introduce an external monitor, critic, verifier, or predictive model for failure detection [4], and rely on large amounts of human takeover data collected during failed executions to learn recovery [5], [6], [7]. This process is costly because operators must continuously monitor rollouts and provide corrective actions across diverse visual, object, and robot-state failures. Moreover, detection

and recovery are learned separately, so a failure alarm is not necessarily aligned with the action needed to recover. We instead seek a single policy that uses endogenous action information to detect and recover from OOD failures without additional failure or intervention data.

DUET takes disagreement between different action spaces as an endogenous signal for failure detection, as illustrated in Fig. 1. In our preliminary experiments, policies trained with relative actions produced smoother and more stable motions during nominal execution, but struggled to recover once a grasp or motion deviated because state-relative commands provide no explicit return target. In contrast, policies trained with absolute actions could redirect the robot toward trajectories supported by the demonstration distribution and retry after a failure. This complementary behavior motivates DUET to learn the same demonstrated behavior with a relative action head and an absolute action head. The relative head serves as the nominal controller, while the absolute head provides global recovery anchors. The two action spaces agree under in-domain execution but necessarily diverge when a failure or OOD event drives the robot away from the demonstrated trajectory, thereby coupling failure detection and recovery through the same signal.

At inference time, DUET converts the predicted absolute waypoint into a correction from the live pose and compares it with the relative action. The two predictions remain consistent along successful trajectories but diverge when a failure drives execution OOD. Their disagreement, together with distance from the training state distribution, determines a soft fusion weight: DUET follows the smooth relative controller during nominal execution, shifts toward the absolute recovery target after failure, and returns to relative control after re-entering the demonstrated support. The same cross-space signal therefore both detects failure and determines the recovery direction, except when the observation itself is invalid and a cached waypoint must be held.

Our contributions are threefold:

- We propose DUET, a dual-action-head VLA that learns relative and absolute action spaces from successful demonstrations and uses their endogenous disagreement to detect OOD-induced failures without an external detector.
- We design a stateless, RVQ-anchored absolute action head and an adaptive closed-loop controller that converts cross-space disagreement into stable recovery while preserving precise relative control under nominal conditions.
- We validate DUET on four LIBERO suites under

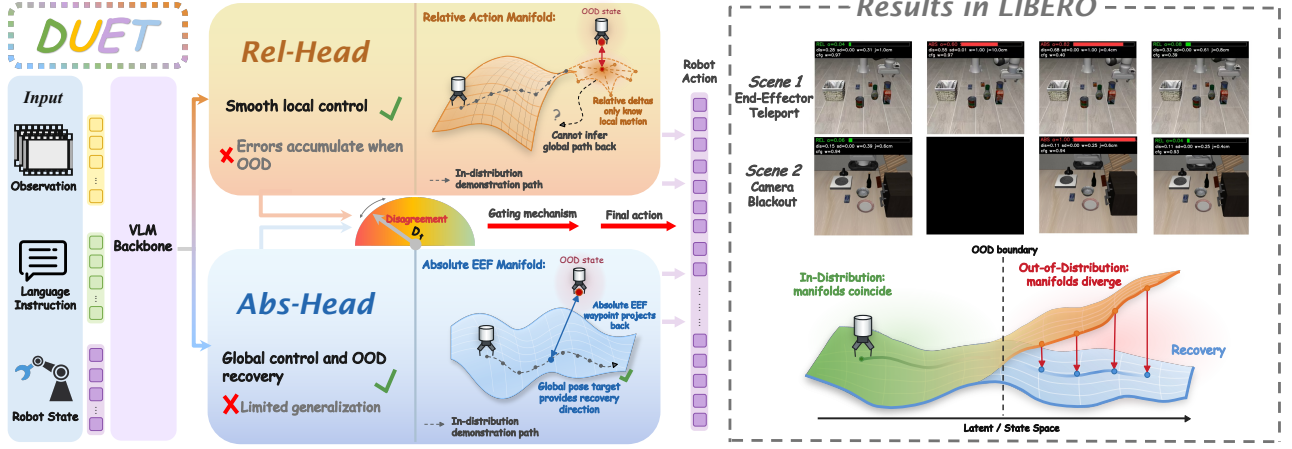


Fig. 1. Dual-manifold failure detection and recovery. The relative and absolute end-effector manifolds agree during nominal execution but diverge when a failure drives the system OOD, exposing an endogenous disagreement score D_t . Relative actions provide precise local motion, whereas absolute waypoints project execution toward a stable demonstrated-manifold anchor.

five failure families—action-execution drift, robot displacement, grasp loss, transient occlusion, and external contact—and transfer the same module to a π_0 backbone on a real Franka arm.

II. RELATED WORK

A. Failure Detection for Robot Policies

Runtime failure detection estimates whether a policy remains within the support of successful behavior. Diff-Dagger uses diffusion uncertainty to request expert intervention [8], while Sentinel monitors action inconsistency and task progress [9]. FAIL-Detect formulates failure detection as sequential OOD testing using only successful executions [4]; FIPER similarly combines representation novelty with action uncertainty without failure data [10].

VLA-specific methods use internal features or external foundation models. SAFE learns a detector from successful and failed rollouts [11], whereas AHA trains a VLM to detect and diagnose failures [12]. Such methods require additional supervision or a separate detector whose output is not a recovery action. DUET instead detects failures through disagreement between two action spaces learned within the policy, using OOD manifold inconsistency as both an alarm and a recovery signal.

B. Recovery from Execution Failures

Recovery methods commonly expand the training distribution with expert corrections. DAgger queries experts on policy-induced states [5]; intervention-based methods collect human takeovers [13], [6]; and IntervGen transforms limited interventions into corrective trajectories [7]. These approaches reduce compounding errors but still require additional recovery actions.

Foundation-model systems instead use semantic supervisors. RACER learns from language-annotated recovery trajectories [14], while AHA and VLM controllers diagnose failures and guide downstream correction [12], [15]. Their

detection and recovery components remain separately trained or prompted. DUET couples both capabilities without extra recovery data: absolute states from successful demonstrations provide recovery anchors, while cross-space disagreement determines when and how strongly to move toward them.

III. METHOD

A. Problem Formulation

At time t , the robot observes multi-view images o_t , a language instruction l , and its current proprioceptive state s_t . DUET is trained only on successful demonstrations and predicts the demonstrated behavior in two action representations over a horizon K :

$$\hat{\Delta}A_t = f_{\text{rel}}(o_t, l, s_t) \in \mathbb{R}^{K \times 7}, \quad (1)$$

$$\hat{W}_t = f_{\text{abs}}(o_t, l) \in \mathbb{R}^{K \times 7}. \quad (2)$$

Here, ΔA_t contains relative operational-space-control commands and W_t contains future absolute end-effector waypoints $[x, y, z, r_x, r_y, r_z, g]$. We decompose each command as $a_t = [a_{p,t}, a_{g,t}]$, where $a_{p,t} \in \mathbb{R}^6$ controls end-effector pose and $a_{g,t} \in \mathbb{R}$ controls the gripper. A successful trajectory naturally supplies both labels: its recorded delta actions supervise f_{rel} , while its future end-effector states supervise f_{abs} . The geometric analysis and fusion operate on a_p ; although the absolute head also predicts gripper state during training, the deployed a_g always comes from the relative head.

B. Dual Action Manifolds

For an instruction l , let $\mathcal{M}_l \subset SE(3)$ denote the end-effector state manifold traced by successful demonstrations, and let TM_l denote its tangent bundle. Under nominal execution, the relative head predicts a local motion vector

$$\hat{a}_{p,t} = f_{\text{rel}}(o_t, l, s_t)_p \in T_{p_t} \mathcal{M}_l, \quad (3)$$

where p_t is the current end-effector pose. This tangent representation is locally smooth and expressive, but it specifies only how to move from the current chart. Once a

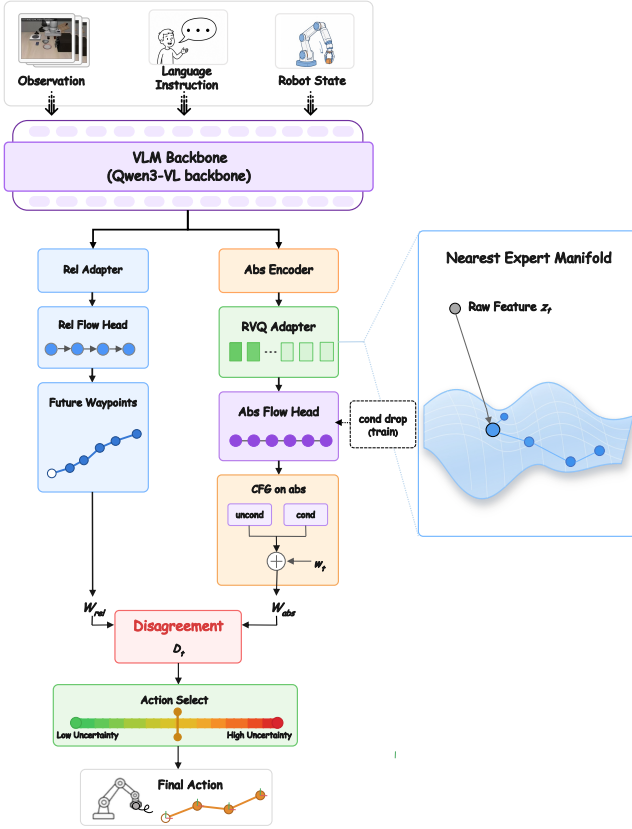


Fig. 2. DUET in four stages: a shared vision-language encoder, relative and absolute action heads, cross-space failure detection, and closed-loop recovery. The relative head provides state-aware local control; the explicitly state-free absolute head uses RVQ-anchored conditions to predict global waypoints. Their disagreement determines the transition between nominal execution and recovery.

failure displaces the robot from the demonstrated support, the extrapolated vector field contains no explicit destination to which the robot should return.

The absolute head instead predicts a future coordinate

$$\hat{w}_{p,t} = f_{\text{abs}}(o_t, l)_p \in \mathcal{M}_l, \quad (4)$$

which acts as a visual-language-conditioned global anchor. To compare the two representations, we introduce a state-dependent transport map

$$c_t = \Psi_{p_t}(\hat{w}_{p,t}) \in T_{p_t}SE(3), \quad (5)$$

which converts the absolute waypoint into a local correction at the live pose. Along an in-domain trajectory, c_t and $\hat{\delta}_{a,p,t}$ describe the same demonstrated motion in two coordinate systems and should therefore align. When a failure drives execution OOD, the relative head extrapolates a local field while the absolute head remains anchored to a demonstrated destination, exposing a cross-space inconsistency.

This geometry couples detection and recovery without asserting a formal OOD guarantee: the norm of the inconsistency provides a scalar failure signal, whereas the transported vector c_t already provides the corresponding recovery direction. DUET can therefore change both when

and where the controller moves using the same pair of predictions.

C. Dual-Head Architecture

Figure 2 shows the shared VLM and two flow-matching action heads [16], [17]. The relative head attends to the full hidden sequence and explicitly receives s_t through a numerical state encoder. It is therefore state-aware and serves as the primary controller during nominal execution. The absolute head receives no explicit current proprioception. When state tokens are appended to the causal VLM prompt, they are placed last and excluded from absolute-branch pooling, while the relative head can still attend to them. This separation prevents the absolute predictor from collapsing into an integration of the current state and preserves its role as a global reference.

The absolute branch pools its state-free hidden sequence H^0 and applies residual vector quantization (RVQ) [18], [19]:

$$Z_q = \text{RVQ}(\text{Pool}(H^0)). \quad (6)$$

RVQ maps continuous conditions to a discrete atlas learned from successful demonstrations, anchoring absolute predictions to recurrent visual-language modes rather than an unconstrained feature vector. Its residual norm is retained only as auxiliary telemetry and is not part of the primary failure score. The absolute flow head additionally uses classifier-free guidance (CFG) [20], enabling its conditional prediction to be softened toward the demonstrated waypoint marginal during recovery.

D. Disagreement-Preserving Learning

The two heads are supervised by the paired labels available in every successful trajectory. Both use flow-matching objectives, while RVQ contributes a commitment loss. The complete default objective is

$$\mathcal{L} = \lambda_{\text{rel}}\mathcal{L}_{\text{rel}} + \lambda_{\text{abs}}\mathcal{L}_{\text{abs}} + \lambda_{\text{vq}}\mathcal{L}_{\text{vq}}, \quad (7)$$

where the absolute labels are normalized future states and the relative labels are the recorded delta commands. We deliberately impose no cross-head consistency loss or distillation: forcing agreement everywhere would suppress the inconsistency needed for failure detection.

DUET instead specializes the branches through asymmetric visual augmentation. The absolute head and RVQ adapter train on both clean and perturbed observations, encouraging the global anchor to remain stable under degraded input. The relative head is optimized only on the clean subset $\mathcal{B}_{\text{clean}}$:

$$\mathcal{L}_{\text{rel}} = \frac{1}{|\mathcal{B}_{\text{clean}}|} \sum_{i \in \mathcal{B}_{\text{clean}}} \ell_{\text{FM}}(\hat{\Delta}^{(i)}, \Delta^{(i)}). \quad (8)$$

This gradient truncation preserves relative-head sensitivity while training the absolute branch as the robust reference, making cross-space disagreement more discriminative under visual OOD. For CFG training, the absolute head also replaces its quantized condition with learned null tokens on a subset of samples and learns both conditional and marginal waypoint flows.

E. Joint Failure Detection and Recovery

At deployment, the policy returns both unnormalized action chunks. Let the live pose be $p_t = (x_t, R_t)$ and the pose component of the current absolute waypoint be $\hat{w}_{p,t} = (\hat{x}_t, \hat{R}_t)$. DUET implements the transport map in Eq. (5) as

$$c_t = \text{clip}\left(\left[\frac{(\hat{x}_t - x_t)/\sigma_p}{\text{Log}(\hat{R}_t R_t^{-1})/\sigma_R}\right], -1, 1\right), \quad (9)$$

where $\text{Log} : SO(3) \rightarrow \mathfrak{so}(3)$ gives the geodesic rotation correction and σ_p, σ_R convert pose error into action units. The cross-space disagreement is

$$D_t = \frac{\|c_t - \text{clip}(\hat{a}_{p,t}, -1, 1)\|_2}{\sqrt{6}}. \quad (10)$$

This is the primary failure signal. We supplement it with a coarse state-distance term that measures normalized violation of the training-state bounds:

$$S_t = \text{mean}\left[\frac{\max(s_{\min} - p_t, 0)}{s_{\max} - s_{\min}} + \frac{\max(p_t - s_{\max}, 0)}{s_{\max} - s_{\min}}\right]. \quad (11)$$

The resulting failure score and recovery weight are

$$q_t = D_t + \lambda S_t, \quad (12)$$

$$\alpha_t = \sigma(k(q_t - \tau)). \quad (13)$$

DUET then interpolates between local execution and global recovery while retaining relative gripper control:

$$a_t = [a_{p,t}, a_{g,t}], \quad (14)$$

$$a_{p,t} = (1 - \alpha_t)\hat{a}_{p,t} + \alpha_t c_t, \quad (15)$$

$$a_{g,t} = \hat{a}_{g,t}, \quad (16)$$

where $[\hat{a}_{p,t}, \hat{a}_{g,t}]$ is the relative-head command. Thus relative control dominates along the demonstrated support, while a rising failure score shifts a_p toward the absolute recovery direction; control returns smoothly after the two representations realign. When requesting a new chunk, the previous fusion weight also softens the absolute-head CFG scale,

$$w_t = w_{\max} - (w_{\max} - w_{\min})\alpha_{t-1}. \quad (17)$$

The disagreement D_t therefore determines *when* recovery is needed, while the vector c_t determines *how* to recover. One exception is treated separately: when the observation is fully invalid, both heads consume the same corrupted input and D_t is uninformative. A validity guard then freezes the last reliable waypoint and holds $\alpha_t=1$ until vision returns.

TABLE I
EXECUTION-FAILURE FAMILIES AND THE QUANTITY DRIVEN OUT OF DISTRIBUTION.

Family	Protocol	What becomes OOD
Action execution	Action perturbation (step 5)	Command-to-motion map
Robot displacement	EEF teleport (step 39)	Robot-arm pose
Object / grasp	Vertical grasp drop (step 67)	Robot-object configuration
Visual occlusion	Camera blackout (step 23)	Current visual observation
Contact / scene	External force	Robot-world interaction

IV. EXPERIMENTS

We organize the simulation experiments around three questions:

- (Q1) Endogenous OOD detection:** Can DUET detect deviations using successful demonstrations only?
- (Q2) Nominal execution:** Does dual-action control preserve normal execution?
- (Q3) Failure-aware retry:** Can a detected, recoverable deviation be converted into task success?

We report detection, nominal success, and closed-loop retry separately, followed by mechanism studies and ablations.

A. Evaluation Protocol

We evaluate the four LIBERO suites [21] with 10 tasks and 50 initial states per task (500 episodes per suite), using suite-specific 10k-step checkpoints. Successful demonstrations provide future end-effector states for the absolute head and delta OSC commands for the relative head.

StarVLA [3] is the external relative-only control. DUET-Rel ($\alpha_t=0$) and DUET share a checkpoint, so their difference isolates adaptive retry; StarVLA→DUET-Rel measures co-training. Absolute-only ($\alpha_t=1$) and fixed fusion ($\alpha_t=0.5$) are inference ablations.

DUET uses Qwen-VL features (size 2048), 16-/8-layer relative/absolute DiT-B heads, and 16 RVQ tokens from three codebooks (size 512, dimension 256). We set $p_{\text{aug}}=0.3$, $p_{\text{drop}}=0.15$, $\lambda=1$, $\tau=0.5$, and $k=8$.

Perturbations use fixed steps, deterministic seeds, and matched (task, initial state) keys; success differences use McNemar p -values. We retain uninjected episodes and report coverage for semantic preconditions. Table I defines the families; external force is reported separately and families are not pooled.

B. (Q1) Endogenous OOD Detection

For **Q1**, Table II compares D_t with matched success-only detectors [4], [9]. Unlike a separate alarm, D_t also activates correction c_t .

With delay close to prior methods (1.41 s versus 1.01–1.31 s), DUET achieves the highest AUROC (0.845) and TPR (0.900). Its balanced accuracy (0.582) remains below RND (0.602), so the gain is not uniform across thresholded metrics. Thus **Q1** is supported: disagreement improves failure ranking and detection without failure labels at modest delay.

TABLE II

FAILURE DETECTION UNDER MATCHED PROTOCOLS. ALL METRICS ARE HIGHER-IS-BETTER EXCEPT DELAY; BOLD MARKS THE BEST VALUE PER COLUMN.

Method	TPR $\uparrow$	TNR $\uparrow$	Bal.-Acc. $\uparrow$	AUROC $\uparrow$	Delay (steps) $\downarrow$	Delay (s) $\downarrow$	Loc.@16 $\uparrow$
DUET (D_t)	0.900	0.265	0.582	0.845	28.3	1.41	0.523
logpZO [4]	0.889	0.243	0.566	0.817	22.7	1.14	0.569
RND [22], [4]	0.842	0.361	0.602	0.549	26.1	1.31	0.330
PCA-kmeans [4]	0.884	0.283	0.583	0.625	20.3	1.01	0.503
STAC [9]	0.839	0.311	0.575	0.494	25.8	1.29	0.260

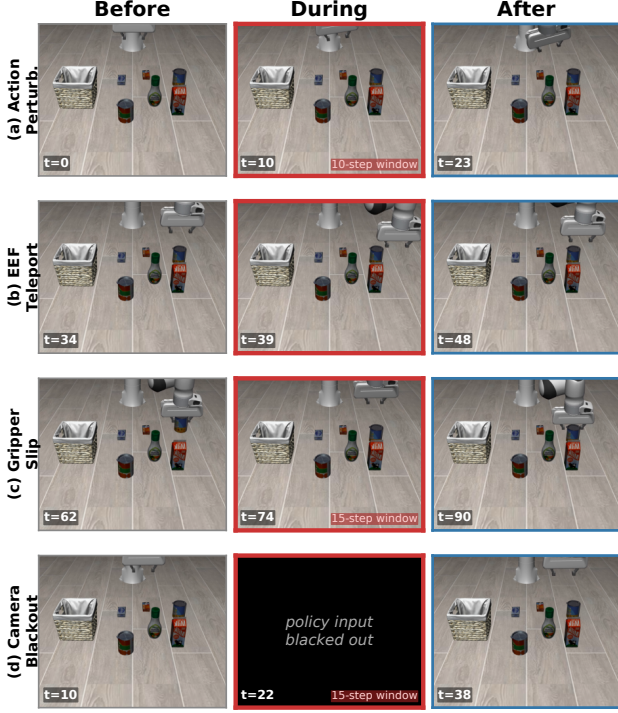


Fig. 3. Object-suite keyframes before, during, and after (a) action perturbation, (b) EEF teleport, (c) grasp drop, and (d) camera blackout.

C. (Q2) Nominal Execution

For **Q2**, adaptive fusion raises the same checkpoint from 93.8% to 96.9% (Table III), with +5.6 pp on Goal and +5.8 pp on LIBERO-10. External rows provide context only. DUET therefore preserves nominal execution while correcting occasional clean-rollout drift.

D. (Q3) Failure-Aware Retry

For **Q3**, Table IV shows that DUET raises shared-checkpoint success from 27.9% to 64.6% under action perturbation, 34.5% to 59.4% after teleport, 89.2% to 95.2% after grasp drop, and 73.5% to 90.8% during blackout. Every aggregate also exceeds StarVLA.

Action execution. The intact scene offers a return target that relative control lacks. DUET gains 36.7 pp over 4,000 pairs; under severe shifts it retains 43.2–60.6%, versus 5.4–12.8% for Rel.

TABLE III

NOMINAL LIBERO SUCCESS (%). EXTERNAL ROWS ARE CONTEXT; STARVLA-GR00T USES JOINT 30K TRAINING AND DUET USES SUITE-SPECIFIC 10K CHECKPOINTS.

Method	Spatial	Object	Goal	LIBERO-10	Avg.
$\pi_{0.5}$ (reported)	98.8	98.2	98.0	92.4	96.9
StarVLA-GR00T (reported)	97.8	98.8	97.4	92.0	96.5
DUET	98.8	99.4	97.6	91.8	96.9
DUET-Rel ($\alpha_t=0$)	98.0	99.2	92.0	86.0	93.8

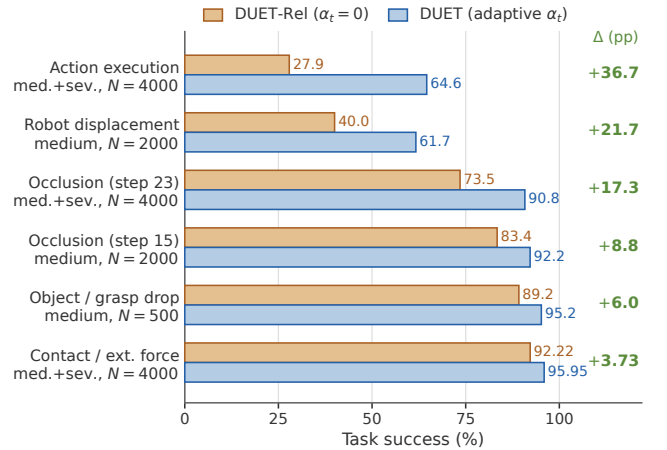


Fig. 4. Paired DUET-Rel→DUET success gain by family.

Robot displacement. Teleport shifts the live pose off-trajectory; DUET gains 24.9 pp overall and 39.8/47.6 pp on Object. Spatial is not significant ($p=0.21/0.067$), partly because episodes can finish before step 39.

Object and grasp. A 5 cm drop changes the robot–object relation but not the arm trajectory. The +6.0 pp gain ($p=2.4 \times 10^{-5}$; coverage $\geq 98.8\%$) over a strong 89.2% Rel baseline makes this a boundary case.

Visual occlusion. A blank image corrupts both heads, so the validity guard, not disagreement, drives recovery. DUET gains 17.3 pp; Object medium is the only regression (−1.8 pp, $p>0.05$), while LIBERO-10 severe gains 60.6 pp.

Contact and scene. External force is separate: DUET reaches 95.95% versus 92.22% over 4,000 episodes, with no suite regression.

a) Mechanism evidence. Figure 5 shows switch-and-return under arm-state OOD: disagreement raises α_t , acti-

TABLE IV

CLOSED-LOOP SUCCESS (%; $N=500$ PER CELL). OVERALL COMBINES MEDIUM+SEVERE ACROSS SUITES. REL→DUET HAS MCNEMAR $p<10^{-3}$ EXCEPT TELEPORT/SPATIAL ($p=0.21/0.067$) AND BLACKOUT/OBJECT MEDIUM.

Suite	Medium				Severe			
	StarVLA	Rel.	DUET	Δ	StarVLA	Rel.	DUET	Δ
<i>(i) Action perturbation (step 5)</i>								
Spatial	60.0	54.0	74.0	+20.0	14.0	12.8	43.2	+30.4
Object	41.6	48.8	84.4	+35.6	7.8	7.8	60.6	+52.8
Goal	52.2	47.2	74.2	+27.0	18.8	10.6	52.6	+42.0
LIBERO-10	36.2	36.6	78.6	+42.0	6.2	5.4	49.0	+43.6
Overall	29.6	27.9	64.6	+36.7	<i>med+sev combined</i>			
<i>(ii) EEf teleport (step 39)</i>								
Spatial	55.4	54.6	57.6	+3.0	46.8	48.2	52.6	+4.4
Object	43.4	36.2	76.0	+39.8	21.2	20.8	68.4	+47.6
Goal	41.6	33.4	52.4	+19.0	30.4	28.2	50.6	+22.4
LIBERO-10	40.4	35.8	60.8	+25.0	21.4	18.4	56.8	+38.4
Overall	37.6	34.5	59.4	+24.9	<i>med+sev combined</i>			
<i>(iii) Vertical grasp drop (Object only, medium, step 67)</i>								
Object	90.6	89.2	95.2	+6.0	—			
<i>(iv) Camera blackout (step 23)</i>								
Spatial	81.0	88.0	94.4	+6.4	42.4	63.2	95.2	+32.0
Object	91.8	98.6	96.8	−1.8	64.2	95.0	97.2	+2.2
Goal	65.4	86.0	86.4	+0.4	35.4	65.8	85.8	+20.0
LIBERO-10	60.0	65.6	84.2	+18.6	24.4	25.6	86.2	+60.6
Overall	58.1	73.5	90.8	+17.3	<i>med+sev combined</i>			

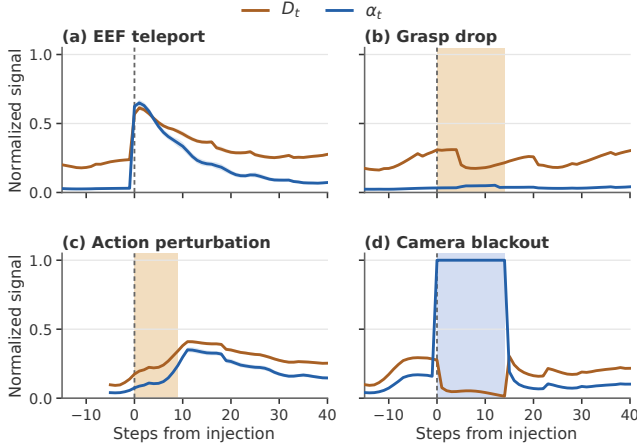


Fig. 5. D_t and α_t on Object, aligned to injection (mean, pointwise 95% CI): arm-state OOD (a,c), world-state change (b), and validity-guard override (d).

vates the waypoint correction, then falls after realignment. The gate stays weak for world-state change and is overridden during complete visual dropout.

On Object teleport (Fig. 6; $n=484-498$), DUET closes 4.69 cm by +32 steps and re-enters in 97.9% (median 16 steps), versus 0.71 cm, 62.5%, and 58 steps for Rel. Absolute-only reaches 99.6% re-entry but zero task success, so re-entry is mechanism evidence rather than the endpoint.

Together, final success and switch-and-return answer **Q3**: DUET helps when deviation is detectable and re-entry remains feasible; grasp changes and invalid vision bound disagreement alone.

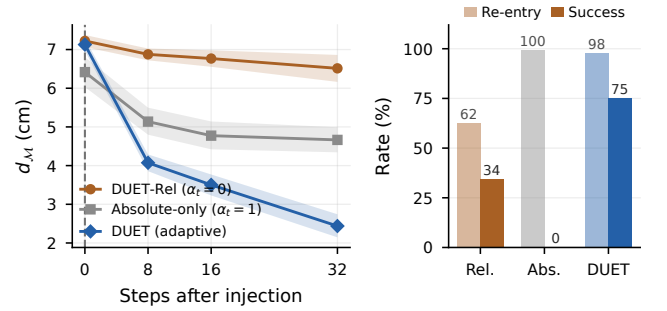


Fig. 6. Object EEf teleport (medium, step 39; $n=484-498$). Left: distance to the clean-trajectory reference after injection. Right: re-entry and final task success.

E. Ablations

Table V tests fusion. Absolute-only loses precise execution; fixed fusion damages clean performance and under-corrects disturbances. Only adaptive fusion succeeds in both regimes.

Table VI isolates other choices. Removing RVQ disables the absolute branch; symmetric augmentation lowers clean and OOD success; adaptive CFG matters off-distribution.

a) Fusion on clean trajectories.: Across 2,000 clean episodes, mean $\alpha_t=0.139$, its 95th percentile is 0.339, and only 1.7% of steps exceed 0.5, showing that relative control remains dominant.

b) Sensitivity to τ and k .: Figure 7 shows a broad Object plateau above 90% clean and 75% OOD success. The shared $\tau=0.5$, $k=8$ lies inside it.

TABLE V
FUSION ABLATIONS, SUCCESS (%). FOUR-SUITE ROWS USE
 $N=2000/2000/4000$; STATIC GATES USE MATCHED OBJECT
PROTOCOLS ($N=500$).

Controller	Gate	Clean	Robot displ.	Action exec.
<i>Four-suite average</i>				
DUET	adaptive α_t	96.9	61.7	64.6
DUET-Rel	$\alpha_t=0$	93.8	40.0	27.9
<i>Object suite, matched protocol</i>				
DUET	adaptive α_t	99.2	76.0	84.4
DUET-Rel	$\alpha_t=0$	99.2	36.2	48.8
Absolute-only	$\alpha_t=1$	5.2	1.1	0.8
Fixed fusion	$\alpha_t=0.5$	46.2	39.4	41.0

TABLE VI
COMPONENT ABLATIONS, SUCCESS (%). CFG USES OBJECT ($N=500$);
RETRAINS USE LIBERO-10 ($N=500/500/1000$).

Variant	Clean	Robot displ.	Action exec.
<i>CFG schedule, Object suite</i>			
Full DUET	99.2	76.0	84.4
No adaptive CFG	88.2	52.6	57.2
<i>Training replacements, LIBERO-10</i>			
Full DUET	91.8	60.8	63.8
RVQ $\rightarrow$ MLP	0.0	0.0	0.0
Symmetric augmentation	86.8	58.6	60.9

F. Real-Robot Evaluation

DUET attaches only to the shared vision-language representation. We replace StarVLA with π_0 (OpenPi) [17], keeping the heads, fusion rule, and validity guard unchanged.

a) Platform and protocol.: A Franka Emika Panda with a wrist-mounted RealSense camera executes four tasks (Fig. 8): pick-and-place, block pushing (non-prehensile), drawer opening (constrained contact), and whiteboard wiping (sustained contact), trained from successful kinesthetic demonstrations only.

Each task is tested under nominal execution and a fixed-magnitude lateral EEf shift injected before grasp, during approach, during handle approach, or during the wiping stroke respectively. For block pushing, the shift deflects the contact point and sends the block off-target. Only trials that receive the injection enter the OOD column; trials are interleaved in randomized blocks and reported as successes/trials with Wilson 95% CIs.

Figure 8 (bottom) shows the switch-and-return behavior. The disturbance matches the robot-displacement family in simulation; Table VII will report whether the π_0 transfer holds.

G. Discussion

The results bound rather than merely confirm the mechanism. D_t ranks failures and drives recovery when the live pose leaves $\mathcal{M}_l$; it is silent on world-state change and uninformative under invalid images (handled by the validity guard). On arm-state OOD, the gate—not co-training—produces the gain (StarVLA $\approx$ DUET-Rel). Under visual

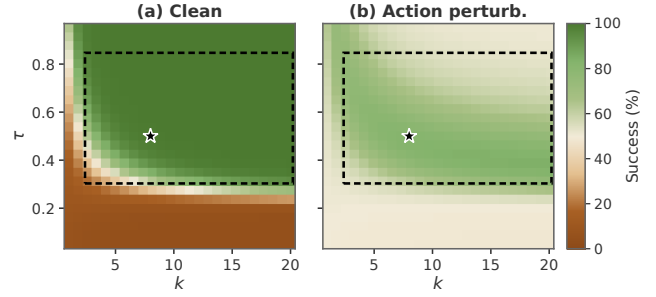


Fig. 7. Object sensitivity to τ, k from 500 clean and 500 perturbed traces. Rectangle: clean $> 90\%$, OOD $> 75\%$; star: operating point.

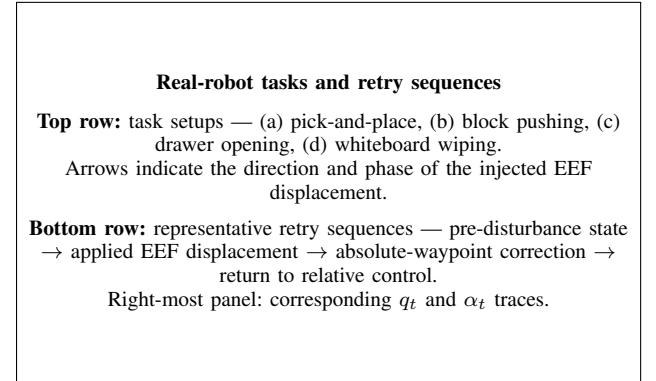


Fig. 8. Franka evaluation. Top: four tasks. Bottom: one retry sequence per task with the corresponding q_t and α_t traces.

dropout the two sources separate: asymmetric training lifts Rel above StarVLA, and the guard adds further improvement on longer outages. The τ - k plateau is broad enough to avoid per-suite tuning.

V. LIMITATIONS

DUET assumes absolute state labels aligned with relative action chunks—natural in LIBERO and many robot logs, but may require calibration on other hardware. The gate uses min-max state statistics as a coarse S_t proxy, which may be insufficient in highly multi-modal workspaces. The stateless absolute branch supports recovery but may lose accuracy when the current robot state carries task information not visible in images. Two scope limits follow: D_t is silent on world-state change (Fig. 5(b)), and both heads degrade together under a fully invalid observation, leaving only the cached-waypoint guard. Irreversible failures and tasks requiring a new strategy are out of scope.

VI. CONCLUSION

We presented DUET, a dual-branch VLA policy for endogenous failure detection and recovery. By aligning a local relative-action tangent field with a stateless, RVQ-anchored absolute-waypoint manifold, DUET uses cross-space disagreement to detect when execution becomes OOD and to activate a closed-loop recovery direction. This unifies failure detection and recovery without an external detector or

TABLE VII

REAL-ROBOT SUCCESS (π_0 BACKBONE), REPORTED AS
 SUCCESSES/TRIALS (%). OOD IS A FIXED-MAGNITUDE LATERAL EEF
 SHIFT INJECTED AT A TASK-SPECIFIC PHASE. Δ IS $+\text{DUET}$ MINUS π_0
 UNDER OOD.

Task	Nominal		OOD (EEF shift)		Δ
	π_0	+DUET	π_0	+DUET	
Pick-and-place (TODO cm)	TODO	TODO	TODO	TODO	TODO
Block pushing (TODO cm)	TODO	TODO	TODO	TODO	TODO
Drawer opening (TODO cm)	TODO	TODO	TODO	TODO	TODO
Whiteboard wiping (TODO cm)	TODO	TODO	TODO	TODO	TODO
Overall	TODO	TODO	TODO	TODO	TODO

additional recovery demonstrations. Because the module attaches only to the shared vision-language representation, the identical design transfers from StarVLA in simulation to π_0 on a real Franka arm without backbone-specific adaptation.

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REFERENCES

- [1] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choro-manski, T. Ding, D. Driess, A. Dubey, C. Finn, P. Florence, C. Fu, M. G. Arenas, K. Gopalakrishnan, K. Han, K. Hausman, A. Herzog, J. Hsu, B. Ichter, A. Irpan, N. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, I. Leal, K.-H. Lee, S. Levine, Y. Lu, H. Michalewski, I. Mordatch, K. Pertsch, K. Rao, K. Reymann, M. Ryoo, G. Salazar, P. Sanketi, P. Sermanet, J. Singh, A. Singh, R. Soricut, H. Tran, V. Vanhoucke, Q. Vuong, A. Wahid, S. Welker, P. Wohlhart, J. Wu, F. Xia, T. Xiao, P. Xu, S. Xu, T. Yu, and B. Zitkovich, “RT-2: Vision-language-action models transfer web knowledge to robotic control,” *arXiv preprint arXiv:2307.15818*, 2023.
- [2] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, T. Lampe, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, C. Finn, and P. Liang, “OpenVLA: An open-source vision-language-action model,” *arXiv preprint arXiv:2406.09246*, 2024.
- [3] StarVLA Community, “StarVLA: A lego-like codebase for vision-language-action model developing,” <https://github.com/starVLA/starVLA>, 2026.
- [4] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez, P. Miller, R. Lee, P. Shah, R. Ambrus, H. Nishimura, and M. Itkina, “Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies,” in *Proceedings of Robotics: Science and Systems*, 2025.
- [5] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, 2011.
- [6] H. Liu, S. Nasiriany, L. Zhang, Z. Bao, and Y. Zhu, “Robot learning on the job: Human-in-the-loop autonomy and learning during deployment,” in *Proceedings of Robotics: Science and Systems*, 2023.
- [7] R. Hoque, A. Mandlekar, C. R. Garrett, K. Goldberg, and D. Fox, “IntervenGen: Interventional data generation for robust and data-efficient robot imitation learning,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2024, pp. 2840–2846.
- [8] S.-W. Lee, X. Kang, and Y.-L. Kuo, “Diff-Dagger: Uncertainty estimation with diffusion policy for robotic manipulation,” in *IEEE International Conference on Robotics and Automation*, 2025.
- [9] C. Agia, R. Sinha, J. Yang, Z. Cao, R. Antonova, M. Pavone, and J. Bohg, “Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress,” in *Proceedings of the Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 689–723.
- [10] R. Römer, A. Kobras, L. Worbis, and A. P. Schoellig, “Failure prediction at runtime for generative robot policies,” in *Advances in Neural Information Processing Systems*, 2025.
- [11] Q. Gu, Y. Ju, S. Sun, I. Gilitschenski, H. Nishimura, M. Itkina, and F. Shkurti, “SAFE: Multitask failure detection for vision-language-action models,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025.
- [12] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang, S. Tian, W. Yuan, R. Krishna, D. Fox, A. Mandlekar, and Y. Guo, “AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation,” in *International Conference on Learning Representations*, 2025.
- [13] J. Spencer, S. Choudhury, M. Barnes, M. Schmittle, M. Chiang, P. Ramadge, and S. Srinivasa, “Learning from interventions: Human-robot interaction as both explicit and implicit feedback,” in *Proceedings of Robotics: Science and Systems*, 2020.
- [14] Y. Dai, J. Lee, N. Fazeli, and J. Chai, “RACER: Rich language-guided failure recovery policies for imitation learning,” in *IEEE International Conference on Robotics and Automation*, 2025, pp. 15 657–15 664.
- [15] H. Chen, Y. Yao, R. Liu, C. Liu, and J. Ichnowski, “Automating robot failure recovery using vision-language models with optimized prompts,” *arXiv preprint arXiv:2409.03966*, 2024.
- [16] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” *The International Journal of Robotics Research*, 2024.
- [17] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, J. Jakubczak, T. Jones, A. Khazatsky, S. Levine, A. Li-Bell, Y. Lu, S. Nair, K. Pertsch, Q. V. Xu, F. Xia, and T. Xiao, “ π_0 : A vision-language-action flow model for general robot control,” *arXiv preprint arXiv:2410.24164*, 2024.
- [18] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, “Neural discrete representation learning,” in *Advances in Neural Information Processing Systems*, 2017.
- [19] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, “Soundstream: An end-to-end neural audio codec,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2022.
- [20] J. Ho and T. Salimans, “Classifier-free diffusion guidance,” *arXiv preprint arXiv:2207.12598*, 2022.
- [21] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, “LIBERO: Benchmarking knowledge transfer for lifelong robot learning,” in *Advances in Neural Information Processing Systems*, 2023.
- [22] Y. Burda, H. Edwards, A. Storkey, and O. Klimov, “Exploration by random network distillation,” in *International Conference on Learning Representations*, 2019.